

# SAT Structural Hardness IIc

From Global Robustness to Multi-Scale Defect-Field Structural Selection

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## Abstract

Papers 50 and 50b identified a limitation of the scalar MAAT compression strategy in SAT hardness benchmarks. Global scores such as  $F_{\text{MAAT}}$  and  $R_{\text{rob}}$  fail to retain the local frustration structure that drives solver effort, while graph-local frustration fields, tail summaries, and hotspot geometry retain useful hardness signal.

This note formulates the corresponding theoretical correction and adds a minimal numerical check on the Paper 50b SAT ensemble. The original global MAAT score is not discarded; it is reinterpreted as a low-frequency projection of a richer defect-field functional. We introduce a multi-scale structural selection form

$$F_{\text{MAAT}}^{\text{multi}} = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}},$$

where mean defects, rare-event tails, connected defect clusters, and scale-dependent defect modes are all allowed to contribute. On the SAT companion benchmark, the components plus density/formula context reach  $R^2 = 0.2833$ , compared with  $R^2 = 0.1375$  for global MAAT scalar/support compression. However, the single re-compressed scalar  $F_{\text{MAAT}}^{\text{multi}}$  remains weak. Thus the lesson is not to replace one scalar by another, but to retain the defect-field components.

In this view, the old scalar form is the macroscopic special case appropriate when defect fields are smooth or self-averaging. SAT, spin-glass-like systems, neural representations, and other frustrated complex systems may require the full defect-field description. The central claim is therefore not that the old formula was wrong, but that it was a macro-projection of a more general multi-scale structural-selection functional.

## 1 Purpose and Status

This is a theoretical companion note to Papers 50 and 50b. It does not introduce a new SAT solver, a new benchmark, or a proof of computational hardness. Its purpose is to update the MAAT formalism in response to the SAT failure:

*global scalar robustness is not the fundamental object in every domain; the more general object is a functional over defect fields.*

The status is deliberately effective-theory level. The proposed functional is a structural selection architecture, not a microscopic derivation from first principles.

## 2 The Problem Exposed by SAT

The earlier scalar support form was

$$F_{\text{MAAT}}[X] = - \sum_a \lambda_a \log(\epsilon + \Gamma_a[X]), \quad \Gamma_a[X] = \frac{1}{1 + d_a[X]}. \quad (1)$$

This form is useful when the relevant structural information is captured by global supports. It worked naturally for smooth field benchmarks, cosmological diagnostics, and other low-frequency structural systems.

SAT exposes a different regime. The compression chain

$$d_a(x) \rightarrow \Gamma_a(x) \rightarrow R_{\text{rob}} \rightarrow F_{\text{MAAT}} \quad (2)$$

can erase the information that matters:

- rare contradictory regions;
- high-frustration tails;
- local propagation traps;
- connected hotspot clusters;
- scale-dependent bottlenecks.

Paper 50b therefore suggests the structural lesson

$$\text{Hardness} \not\sim f(\text{global robustness}), \quad \text{Hardness} \sim f(\text{multi-scale frustration geometry}). \quad (3)$$

### 3 Defect Fields

Let  $G$  be a graph, lattice, spacetime region, computational trajectory, or other structured domain. Instead of beginning with one global defect  $d_a[X]$ , define local defect fields

$$d_a(x), \quad x \in G, \quad (4)$$

and corresponding local supports

$$\Gamma_a(x) = \frac{1}{1 + d_a(x)}. \quad (5)$$

The local fields may represent different structures depending on the domain:

| Domain              | Relevant defect-field structure                                    |
|---------------------|--------------------------------------------------------------------|
| Smooth field theory | local residuals, gradients, stability margins                      |
| Cosmology           | background sectors, growth sectors, projection kernels             |
| SAT / constraints   | propagation traps, contradiction fields, backdoor geometry         |
| Spin glasses        | frustrated plaquettes and cluster structure                        |
| Neural systems      | representation bottlenecks and unstable activation regions         |
| String landscapes   | basin geometry, transition obstructions, local EFT-control defects |

The global scalar theory is recovered only after choosing a projection from the field  $d_a(x)$  to a finite set of summary observables.

## 4 Multi-Scale Structural Functional

We propose the effective multi-scale form

$$\boxed{F_{\text{MAAT}}^{\text{multi}}[X] = F_{\text{mean}}[X] + F_{\text{tail}}[X] + F_{\text{cluster}}[X] + F_{\text{scale}}[X].} \quad (6)$$

The four terms are:

$$F_{\text{mean}} = \sum_a \lambda_a^{(0)} \langle d_a(x) \rangle_G, \quad (7)$$

$$F_{\text{tail}} = \sum_a \lambda_a^{\text{tail}} q_p(d_a), \quad p \in \{0.90, 0.95, 0.99\}, \quad (8)$$

$$F_{\text{cluster}} = \sum_a \lambda_a^{\text{cl}} \frac{C_{\text{max}}^{(a)}}{|G|}, \quad (9)$$

$$F_{\text{scale}} = \sum_{a,k} \lambda_{a,k} d_a^{(k)}. \quad (10)$$

Here  $q_p(d_a)$  is a high quantile of the defect field,  $C_{\text{max}}^{(a)}$  is the largest connected high-defect cluster, and  $d_a^{(k)}$  denotes a scale-resolved mode, shell, or coarse-grained component.

The associated selection measure is

$$\boxed{P[X] = \frac{1}{Z} \exp[-F_{\text{MAAT}}^{\text{multi}}[X]].} \quad (11)$$

## 5 SAT Specialisation

For a SAT formula with clause-variable graph  $G$ , a concrete field-level functional can be written as

$$\boxed{F_{\text{MAAT}}^{\text{field}}[G] = \sum_{x \in G} \sum_a \lambda_a(x) d_a(x) + \lambda_{\text{tail}} q_{95}(d_{\text{frust}}) + \lambda_{\text{cluster}} \frac{C_{\text{max}}}{n} + \lambda_{\text{var}} \text{Var}(d_{\text{frust}}).} \quad (12)$$

The local frustration field may include degree anomalies, polarity imbalance, short-cycle density, propagation depth, contradiction rate, and backdoor-density participation.

This form preserves exactly the structures that scalar  $F_{\text{MAAT}}$  can destroy: rare peaks, heavy tails, hotspot clusters, and local bottlenecks.

## 6 Relation to the Old Formula

The old formula is not rejected. It becomes the low-frequency projection:

$$F_{\text{MAAT}}^{\text{old}} \simeq F_{\text{mean}}, \quad (13)$$

or, more generally,

$$F_{\text{MAAT}}^{\text{old}} = \Pi_{\text{macro}} \left[ F_{\text{MAAT}}^{\text{multi}} \right], \quad (14)$$

where  $\Pi_{\text{macro}}$  is a macroscopic projection that discards tails, clusters, and scale-resolved modes.

This distinction explains why the old form can work in smooth systems but fail in SAT-like systems:

| Regime                     | Defect structure         | Appropriate MAAT level                |
|----------------------------|--------------------------|---------------------------------------|
| Smooth / self-averaging    | low variance, weak tails | scalar global $F_{\text{MAAT}}$       |
| Critical / intermittent    | tails and bursts         | mean + tail functional                |
| Clustered / frustrated     | connected defect islands | mean + tail + cluster functional      |
| Hierarchical / multi-scale | scale-dependent defects  | full $F_{\text{MAAT}}^{\text{multi}}$ |

## 7 Response Closure

The response-based closure of weights can also be lifted from scalars to defect-field observables. Let

$$\mathcal{O}[X] = \left( \langle d_a \rangle, q_p(d_a), C_{\max}^{(a)} / |G|, d_a^{(k)} \right) \quad (15)$$

collect all selected structural observables. Then the linear-response closure becomes

$$\lambda_{\mathcal{O}} = \left( C_{\mathcal{O}} + \eta \frac{\text{tr}(C_{\mathcal{O}})}{M} \mathbb{I} \right)^{-1} (\langle \mathcal{O} \rangle_{\mu_0} - \langle \mathcal{O} \rangle_*), \quad (16)$$

where  $M$  is the number of retained structural observables. Thus response closure is not restricted to five global sector weights. It can act on an expanded observable basis containing mean, tail, cluster, and scale modes.

## 8 Numerical Check on Paper 50b

To test whether the multi-scale functional is more than notation, we added the four components

$$F_{\text{mean}}, \quad F_{\text{tail}}, \quad F_{\text{cluster}}, \quad F_{\text{scale}}$$

to the Paper 50b SAT experiment. The implemented scalar is

$$F_{\text{MAAT}}^{\text{multi}} = F_{\text{mean}} + F_{\text{tail}} + F_{\text{cluster}} + F_{\text{scale}}. \quad (17)$$

The benchmark is unchanged otherwise: 550 synthetic random, planted, and modular 3-SAT instances are evaluated against the deterministic DPLL node-count target. The result is intentionally reported as an ablation, not as a new SAT solver.

| Random-forest model                                           | $R^2$         |
|---------------------------------------------------------------|---------------|
| Density-only                                                  | 0.0362        |
| Density plus formula type                                     | 0.1834        |
| Standard graph                                                | 0.2959        |
| Global MAAT scalar/supports                                   | 0.1375        |
| Single scalar $F_{\text{MAAT}}^{\text{multi}}$                | -0.0806       |
| Components of $F_{\text{MAAT}}^{\text{multi}}$                | -0.0224       |
| $F_{\text{MAAT}}^{\text{multi}}$ components plus density/type | 0.2833        |
| Local frustration plus density/type                           | 0.2946        |
| Raw graph plus frustration                                    | <b>0.3152</b> |

This check refines the theoretical claim. The multi-scale components are useful once the phase-density and formula-family context is retained, improving over global MAAT by

$$\Delta R^2 = 0.2833 - 0.1375 = 0.1458.$$

But the single scalar  $F_{\text{MAAT}}^{\text{multi}}$  fails. Therefore the update should not be read as “replace  $F_{\text{MAAT}}$  by a better scalar”. It should be read as:

*keep the defect-field resolution; do not immediately collapse it back to one number.*

## 9 Interpretation

The conceptual shift is:

| Old interpretation             | Updated interpretation                             |
|--------------------------------|----------------------------------------------------|
| Global structural robustness   | low-frequency structural projection                |
| Five scalar supports           | projected summaries of defect fields               |
| One score explains all domains | each domain requires appropriate resolution        |
| SAT failure is a contradiction | SAT failure diagnoses missing high-frequency modes |

The central sentence is:

*Global MAAT is the macro projection; full MAAT is the defect-field functional.*

The numerical check strengthens this sentence but also constrains it: field-level components carry more signal than the old scalar, while immediate re-compression into a new scalar again loses information.

## 10 Limitations

The proposed functional is still phenomenological. Several questions remain open:

- how to derive the correct defect fields from microscopic principles;
- how to choose quantile, cluster, and scale observables without overfitting;
- how to compare field-level MAAT against strong graph-learning baselines;
- how to calibrate the enlarged response closure without parameter soup;
- how to identify domains where the scalar macro projection is sufficient.
- how to test the field-level functional against modern CDCL targets and external SAT benchmarks.

## 11 Conclusion

Papers 50 and 50b do not invalidate MAAT. They identify the resolution at which the current scalar theory fails. For SAT, the relevant structures are local, rare, clustered, and multi-scale. This motivates the v1.4 update:

MAAT v1.4: multi-scale defect-field structural selection.

The old scalar  $F_{\text{MAAT}}$  remains valid as a macro projection in smooth or self-averaging systems. The full theory should be understood as a functional over defect fields, and the SAT test indicates that those field components should often be retained rather than immediately re-compressed into a single number.

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